SAY WATT ROBOTICS · FIELD LOG

Week 2.3 — Two ESP32s Start Talking — A First Radio Link, a First Sensor, and a Wire That Wouldn't Fit

Sunday, October 4, 2026 · 3:00 – 5:00 PM · 9 students

Nine of us at 2.3 — Rhylan, Tyler, Trenton, Sam, Frankie, Luke, Oliver, Eli, and Maddox — from 3:00 to 5:00. The profiler went from a list to parts on the table. Management turned the bill of materials into a parts list and updated the budget to match. The software track got two ESP32 microcontrollers talking to each other over ESP-NOW, a protocol that lets the boards send data straight to one another with no Wi-Fi network in between, then wired a temperature sensor into the transmitter board and tested it in air — all of it written in C++ in the Arduino IDE. Mechanical sharpened the sizing math: in a four-inch pipe, every centimeter of height adds water pushed aside and adds pipe weight, and the difference between the two is how much lift each centimeter buys. They also measured four- and five-inch circles to see how many syringes each diameter could hold, and prototyping started the cheapest way there is — with cardboard. Electrical soldered the power rail box and finished the 12-volt terminal block, then ran into the wall every electrical team hits eventually: the 18-gauge wire would not fit through the holes on the buck converter that steps 12 volts down to 5, so that joint waits for next week. The payload track went back over the manipulator trade-offs. And the team is looking for a new marketing lead.

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STUDENTS AT THE SESSION	ESP32S TALKING OVER ESP-NOW	SENSOR WIRED INTO THE FLOAT ELECTRONICS	WIRE TOO THICK FOR THE BUCK CONVERTER'S HOLES

WHAT WE LEARNED

- ESP-NOW lets two ESP32s talk directly — no router, no Wi-Fi network, one board sending to the other by its hardware address. That matters at a competition where other teams' floats will be transmitting at the same time.
- Wire gauge has to fit the hardware. 18-gauge wire carries current well and is useless if it won't go through the hole on the board. That is a five-second check that saves an afternoon.
- Every centimeter of a float does two things at once: it pushes more water aside, which adds lift, and it adds more pipe, which adds weight. Net buoyancy per centimeter is the difference, and it lets you size the float with arithmetic.
- Measure the circle before you design the bundle. How many syringes fit inside a four-inch tube versus a five-inch tube decides how much water the buoyancy engine can move.
- Cardboard is the cheapest prototype there is. A syringe layout cut from a box tests the idea in minutes for nothing.
- A sensor in air is a start, not a result. The temperature sensor works on the bench; the float has to read water to within two degrees Celsius at four meters.

DECISIONS MADE

- ESP-NOW is the first communication method proven on the bench. Of the options on the table — Bluetooth, Wi-Fi, and ESP-NOW — it is the one now working board to board.
- The float is programmed in C++ in the Arduino IDE, on both the transmitter and the receiver.
- Height is the tuning knob for buoyancy. In a four-inch pipe each centimeter of height adds a predictable amount of net lift, which turns the float's length into a calculation instead of a guess.
- Prototype in cardboard first. Syringe layouts get cut out of cardboard before anything is printed or bought.

ACTION ITEMS

- Get the parts in and inventory them
  Sensor group · by 2.4
- Test SD card storage on the ESP32
  Software group · by 2.4
- Finish the 12 V to 5 V buck converter connection
  Electrical group · by 2.4
- Find a new marketing lead
  Management · by 2.4

STUDENT LEARNING OUTCOMES

- Established ESP-NOW communication between two ESP32 microcontrollers.
- Integrated a temperature sensor with the ESP32 transmitter and tested it in air, programming both boards in C++ in the Arduino IDE.
- Calculated how a float's net buoyancy changes with its height in a four-inch pipe.
- Measured four- and five-inch diameters to determine how many syringes each could hold.
- Soldered the power rail box and completed the 12-volt terminal block.
- Diagnosed a wire-gauge mismatch at the buck converter.
- Turned the bill of materials into a parts list and updated the season budget.
- Started cardboard prototyping of the profiler.

LOOKING AHEAD

Parts come in and get inventoried. The software track adds SD card storage so the float can hold its data until it surfaces, electrical goes back to the buck converter with a wire that fits, and the search for a new marketing lead continues.

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